

[Pranesh S A 251801234-2025-AIDS](#) ▾**P2****Started on** Friday, 4 September 2026, 7:58 PM**State** Finished**Completed on** Friday, 4 September 2026, 8:07 PM**Time taken** 8 mins 20 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, x;
5      if (scanf("%d", &n) != 1) return 0;
6
7      int arr[n];
8      for (int i = 0; i < n; i++) {
9          scanf("%d", &arr[i]);
10     }
11     scanf("%d", &x);
12
13     int low = 0, high = n - 1;
14     int ans = -1;
15
16     while (low <= high) {
17         int mid = low + (high - low) / 2;
18         if (arr[mid] <= x) {
19             ans = arr[mid];
20             low = mid + 1;
21         } else {
22             high = mid - 1;
23         }
24     }
25
26     printf("%d", ans);
27     return 0;
28 }
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			

	Input	Expected	Got	
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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